

REQUEST FOR PROPOSAL - DEMO PACKAGE

Refinery Severe-Service Control Valve Package

Resid Hydroprocessing Unit Control Valve Replacement

FICTIONAL CUSTOMER PROFILE

NorthStar Refining and Petrochemicals Complex

INDUSTRY

Refinery / Petrochemical

ISSUE DATE

29 May 2026

BID DUE DATE

24 June 2026

COMMERCIAL CONTEXT

Estimated package value: INR 21.8 Cr

PACKAGE TYPE

Severe-service control valve proposal

Demo-safe fictional RFP package created for WinsProposal. Customer names are fictional industry-style accounts and do not represent official documents, endorsements, trademarks, or partnerships.

1. Purchaser Profile and Project Background

NorthStar Refining and Petrochemicals Complex is a fictional refinery major profile used to represent a sophisticated process owner with strong QA/QC, NACE, inspection, and technical bid evaluation expectations.

The purchaser seeks severe-service control valves for high differential pressure refinery liquid services with potential cavitation, flashing, erosion, corrosion, and sour-service exposure. The package must include a strong proposal-stage engineering narrative, material compatibility notes, deviation handling, and a complete inspection/testing plan.

Proposal expectation: Supplier response shall show engineering proposal intelligence: extracted requirements, compliance/TBE readiness, severe-service risk recognition, knowledge reuse, deviations, review owners, and management-ready documentation.

2. Scope of Supply

- Supply severe-service refinery control valves with tag-wise technical compliance.
- Provide cavitation/flashing/choked-flow narrative and preliminary engineering review checklist.
- Provide NACE MR0175 / ISO 15156 applicability matrix by tag where sour service is confirmed.
- Provide QA/QC, hydrotest, PMI, material certificate, and deviation register documentation.
- Support technical bid evaluation with material, trim, pressure class, actuator, testing, inspection, and engineering comments.

3. Line Items

Tag / Ref.	Item	Qty	Indicative Size / Class	Service Notes
FV-RFU-2201A/B	High-dP refinery letdown control valve	2	NPS 6, ASME 600	Potential cavitation
LV-RFU-2240A/B/C	Sour hydrocarbon level control valve	3	NPS 4, ASME 600	NACE applicability to be confirmed
PV-RFU-2288	Flashing service pressure control valve	1	NPS 3, ASME 900	Flashing/erosion risk
DOC-RFU	QA/QC and engineering dossier	1 lot	MDR package	Compliance, deviations, ITP, certificates

4. Process Conditions

Input	Requirement / Value
Fluid	Sour hydrocarbon / refinery process liquid
Inlet pressure	76 barg
Outlet pressure	22 barg
Temperature	180 C
Vapor pressure	To be confirmed by supplier clarification
Service concerns	Cavitation, flashing, erosion/corrosion, sour-service material restrictions
Leakage class	Class IV/V by tag

5. Technical Specification

- Supplier shall classify each tag for cavitation, flashing, erosion, corrosion, sour-service, noise, and vibration risk.
- Supplier shall recommend anti-cavitation trim or flashing-resistant configuration where applicable.
- Supplier shall identify missing fluid properties required for final sizing and shall not claim certified final Cv/Kv without complete data.
- Supplier shall state pressure class, body material, trim material, hardfacing, packing, bolting, and NACE assumptions.
- Supplier shall include deviation handling for material substitutions, trim alternatives, testing exceptions, and documentation assumptions.
- Supplier shall include engineering validation checklist for final sizing, material selection, leakage class, and inspection scope.

6. Compliance Requirements

- ISA 75.01 / IEC 60534 for control valve sizing awareness.
- NACE MR0175 / ISO 15156 for sour-service material requirements where applicable.
- ASME B16.34 for pressure-temperature considerations.
- Project hydrotest, seat leakage, PMI, NDE, and material traceability requirements.
- Client QA/QC procedure and manufacturing data record requirements.

7. Inspection and Testing Requirements

- Hydrostatic shell test and seat leakage test.
- PMI for pressure-retaining and trim components where specified.
- NDE per line class or project inspection category.
- MTC review and traceability verification.
- Client witness point for hydrotest and final inspection.
- Final data book with approved deviation register.

8. Required Proposal-Stage Engineering Intelligence

Engineering Narrative Required

- Extract process conditions and classify application/service.
- Provide preliminary valve configuration narrative.
- Provide severe-service risk assessment.
- Provide cavitation/flashing/choked-flow narrative where relevant.
- Provide noise/vibration risk narrative.
- Provide trim, material, actuator, and accessory notes.

Calculation Summary Required

- State process inputs and assumptions.
- Classify pressure drop severity.
- Provide indicative Cv/Kv placeholder only where safe.
- List risk flags and missing data.
- List validation checklist and standards awareness.
- State that final sizing/design requires engineer validation.

9. Technical Bid Evaluation Template

Evaluation Tag	Purchaser Requirement
Material	Supplier to map material by tag with NACE note where applicable.
Trim	Anti-cavitation or flashing-resistant trim to be proposed based on validated conditions.
Pressure class	ASME class to match refinery line class and ASME B16.34 review.
Actuator	Actuator sized for shutoff, fail action, leakage class, and site instrument air.
Testing	Hydrotest, seat leakage, PMI, NDE where required, and witness points.
Deviation	All exceptions to be listed with technical, commercial, and schedule impact.

10. Evaluation Criteria

Criterion	Weight
Technical compliance	35%
Severe-service risk response	25%
QA/QC and documentation	15%
Delivery	10%
Commercial value	15%

11. Possible Deviations and Clarifications

- Vapor pressure and fluid properties are incomplete for final cavitation/flashing validation.
- Sour-service tag list and NACE environmental limits require client confirmation.
- Hardfacing and exotic metallurgy shall be priced subject to validated erosion/corrosion basis.

12. Engineering Review Points

- Confirm vapor pressure, density, viscosity, critical pressure, and maximum/minimum flow cases.
- Confirm NACE sour-service tags, H₂S partial pressure, chlorides, pH, and material limits.
- Confirm whether service is cavitating, flashing, or clean liquid pressure letdown.
- Confirm leakage class by tag and shutoff pressure.
- Confirm client acceptance process for technical deviations.

13. Commercial and Delivery Assumptions

- Supplier shall state drawing submission lead time, manufacturing lead time, test duration, and shipping readiness.
- Supplier shall identify long-lead materials, special trims, actuator accessories, and inspection hold points that may affect schedule.
- Supplier shall provide itemized pricing by line item and list optional compliance or testing costs separately.
- Supplier shall include a deviation register with technical, commercial, and schedule impact.
- Proposal validity shall be not less than 60 days from bid due date.

Mandatory wording: Any engineering analysis in the supplier proposal shall clearly state: "Preliminary proposal-stage engineering estimate. Final sizing/design must be validated by qualified engineers using company-approved tools and standards."